

ADVANCED RENEWABLE ENERGY SYSTEMS, DECARBONIZATION & CLIMATE ECONOMICS

New Egyptian Baccalaureate Curriculum (Stage 3)

ACADEMIC YEAR: 2024 / 2025 | BACCALAUREATE STAGE 3

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Advanced Egyptian Baccalaureate textbook in Renewable Energy Systems & Climate Economics: Perovskite-silicon tandem solar cell physics, blade element momentum (BEM) offshore wind aeroelasticity, Faraday efficiency in high-temperature SOEC electrolysis, trans-Mediterranean HVDC grid interconnectors, and econometric marginal abatement cost curves (MACC) with EU CBAM compliance.

CORE CURRICULUM HIGHLIGHTS:

- Shockley-Queisser theoretical efficiency limit and multi-junction perovskite bandgap tuning
- Blade Element Momentum (BEM) theory, wake aerodynamic interactions, and deepwater floating spar-buoy dynamics
- Electrochemical overpotential polarization curves, electrocatalytic Pt-Ir catalysts, and Fischer-Tropsch e-fuels
- Synchronous virtual inertia, HVDC voltage source converter (VSC) topology, and corporate EU CBAM carbon tariffs

APPROVED CHAPTER MODULES:

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|---|----------------------------|
| - Next-Gen Photovoltaic Semiconductor Physics & Utility Solar Farms | pp. 1 74 (4 core units) |
| - Advanced Wind Aerodynamics, Offshore Wind Farms & Aeroelasticity | pp. 75 148 (4 core units) |
| - Green Hydrogen Thermodynamics, PtX Catalysis & Maritime Decarbonization | pp. 149 222 (4 core units) |
| - Grid Flexibility, Interconnection Markets & Industrial CBAM Compliance | pp. 223 296 (4 core units) |

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Advanced Clean Energy Systems, PtX Catalysis & Climate Economics Compendium
Full Ministerial Textbook: 296 Pages - 34.5 MB (Direct on MOE Portal)
Publisher: Supreme Council for Baccalaureate Education & Ministry of Education
Official Portal: <https://moe.gov.eg/ar/elearning-content/> - Code: EGBAC-RENEW-2025
Legal Deposit: 2025/14892 - ISBN: 978-977-12-4852-95



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Advanced Egyptian Baccalaureate textbook in Renewable Energy Systems & Climate Economics: Perovskite-silicon tandem solar cell physics, blade element momentum (BEM) offshore wind aeroelasticity, Faraday efficiency in high-temperature SOEC electrolysis, trans-Mediterranean HVDC grid interconnectors, and econometric marginal abatement cost curves (MACC) with EU CBAM compliance. Shockley-Queisser theoretical efficiency limit and multi-junction perovskite bandgap tuning Blade Element Momentum (BEM) theory, wake aerodynamic interactions, and deepwater floating spar-buoy dynamics Electrochemical overpotential polarization curves, electrocatalytic Pt-Ir catalysts, and Fischer-Tropsch e-fuels Synchronous virtual inertia, HVDC voltage source converter (VSC) topology, and corporate EU CBAM carbon tariffs

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	45% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: <https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Next-Gen Photovoltaic Semiconductor Physics & Utility Solar Farms	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Advanced Wind Aerodynamics, Offshore Wind Farms & Aeroelasticity	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Green Hydrogen Thermodynamics, PtX Catalysis & Maritime Decarbonization	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Grid Flexibility, Interconnection Markets & Industrial CBAM Compliance	7 wks	15 pts	25.0%	4 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: NEXT-GEN PHOTOVOLTAIC SEMICONDUCTOR PHYSICS & UTILIZATION

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Next-Gen Photovoltaic Semiconductor Physics & Utility Solar Farms.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Next-Gen Photovoltaic Semiconductor Physics & Utility Solar Farms.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: ADVANCED WIND AERODYNAMICS, OFFSHORE WIND FARMS & AEROELASTICITY

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Advanced Wind Aerodynamics, Offshore Wind Farms & Aeroelasticity.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Advanced Wind Aerodynamics, Offshore Wind Farms & Aeroelasticity.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: GREEN HYDROGEN THERMODYNAMICS, PTX CATALYSIS & MARITIME DECARBONIZATION

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Green Hydrogen Thermodynamics, PtX Catalysis & Maritime Decarbonization.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Green Hydrogen Thermodynamics, PtX Catalysis & Maritime Decarbonization.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

ADVANCED RENEWABLE ENERGY SYSTEMS, DECARBONIZATION & CLIMATE ECONOMICS
Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion: $\sum F_{\text{net}} = 0, \sum M_{\text{pivot}} = 0$
Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \partial \rho / \partial t = 0$
Normalized Standard Score: $Z = (X - \mu) / \sigma$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES
Linear Operator Transformation: $T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$
Sample Variance Boundary: $s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio: $\eta = (\text{Output}_{\text{actual}} / \text{Input}_{\text{standard}}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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